

Predicting how chaperone binding responds to salt, heat and staurosporine from protein sequence

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Summary

Recombinant, tagged chaperones were added to a human cell lysate and pulled down with the proteins they bind, under different conditions applied to the lysate: salt (HSPB1), and temperature and the kinase inhibitor staurosporine (DNAJA1 and DNAJB11). Mass spectrometry measured how much of each protein each pull-down contained. The question is how well a protein's sequence predicts how its binding changes, and what the predictions reveal.

Response	Bait	Ceiling	Sequence	Composition
Co-chaperone preference	DNAJB11 vs DNAJA1	96 %	49 %	32 %
Heat, 43 vs 35 °C	DNAJA1, DNAJB11	99 %	27 %	9 %
Salt, 150 vs 0 mM	HSPB1	99 %	20 %	8 %
Salt, 75 vs 0 mM	HSPB1	99 %	18 %	6 %
Heat, 37 vs 35 °C	DNAJA1, DNAJB11	87 %	13 %	4 %
Staurosporine, 37 °C	DNAJA1, DNAJB11	76 %	7 %	0 %

Percent of variance explained on proteins whose sequence families were held out of training, by the best sequence model and by the 20 amino-acid fractions alone. The ceiling is the part of the variance that is reproducible between independent replicates, and so the most any predictor could explain.

- **Sequence predicts every response, well beyond amino-acid composition.** The best results come from reading several layers of the ESMC-6B protein language model at once.
- **Each response has its own, interpretable determinants.** Salt spares proteins with transmembrane helices. Heat recruits folded cytoplasmic proteins and loses disordered and membrane proteins. Staurosporine removes kinases, in proportion to how tightly it binds them. DNAJB11 prefers secretory and membrane proteins, DNAJA1 nuclear proteins with charged disordered regions.
- **Thermal stability sets when a protein's binding switches on.** The least stable proteins respond by 37 °C, those melting a few degrees above 43 °C respond between 37 and 43 °C, and the most stable barely respond; comparing 43 with 35 °C therefore gives an inverted U. Stability matters, but explains little of the response overall.
- **The responses are largely independent of one another.** There is no single protein property behind them all.

1. The experiments

Design. All manipulations were made in vitro. A tagged chaperone was added to cell lysate, incubated under the condition of interest, and recovered with its bound proteins.

Experiment	Bait	Conditions (in lysate)	Replicates	Proteins
Salt titration	HSPB1	0, 75, 150 mM salt	5 x 2 technical	7,184
GA_33	DNAJA1, DNAJB11	35/37/43 °C, +/- 10 µM staurosporine	6	9,871

Because everything happens in lysate, there is no cellular stress response, synthesis, degradation or relocalisation: a change in a pull-down reflects a change in binding, or in how much of a protein stays soluble during the incubation. The two co-chaperones meet the same mixture of proteins, so any difference between them is a difference in what they bind, not in where they normally live (DNAJB11 is an ER-luminal protein, DNAJA1 a cytosolic and nuclear one).

What is measured. For each protein, the change in its log2 intensity in the pull-down between two conditions (a log2 fold change), averaged over biological replicates. A protein is kept when it is detected in most replicates of both conditions. GA_33 runs are median-normalised; the main conclusions are unchanged under four other normalisations. In GA_33 the heat and staurosporine responses are averaged over the two co-chaperones, which respond almost identically (correlations 0.83-1.0 after correcting for noise).

Measurement ceiling. Each fold change was recomputed from two disjoint halves of the biological replicates; their agreement, corrected to the full replicate count, is the reliability. It is 99 % for salt and for heat at 43 °C, so these responses are measured almost without noise. Staurosporine changes binding only about 1.15-fold, and a quarter of its variance is replicate noise (ceiling 72-76 %).

2. Predicting a response from sequence

Each protein’s sequence is embedded with ESMC-6B, a protein language model, and averaged over residues. Ridge regression predicts the response from the embedding. Proteins are grouped into families at 30 % sequence identity, and every model is scored on families it has never seen, so a prediction is never made from a close relative. Intervals resample those families.

- **Composition first.** The 20 amino-acid fractions explain 6-9 % of the salt and heat responses and 32 % of the co-chaperone preference. Charge, hydrophobicity and charge patterning add almost nothing beyond composition. The embedding explains 2-3 times as much, and contains composition.
- **Several layers beat one.** The most predictive layer differs by response (the final layer for heat and the preference, layer 50 for salt). Combining nine layers with learned weights beats the best single layer on every response, by 0.7 to 3 points, and gives the figures in the summary table.
- **Linear is nearly enough.** A small neural network averaged with ridge adds 1-2 points for heat and salt; it does not help for staurosporine, whose signal sits in a few kinase families.
- **Sparse-autoencoder features make the predictions readable.** Biohub’s sparse autoencoder for ESMC-6B (layer 60) decomposes each residue into 64 active features out of 16,384, each with an annotated meaning. These features keep 90-114 % of the embedding’s predictive power, and the features that predict each response are reported below. Each label was checked against the annotations of the proteins it actually fires on in these data.

3. Salt

Salt weakens electrostatic interactions and strengthens hydrophobic ones. A typical protein’s HSPB1 binding changes 1.6-fold at 75 mM and 1.9-fold at 150 mM.

Features (each row adds to the one above)	75 mM	150 mM
Protein length	0.0 %	0.0 %
Amino-acid composition	6.2 %	7.5 %
+ charge, hydrophobicity, charge patterning	6.6 %	8.2 %
ESMC embedding, one layer	16.0 %	18.2 %

Features (each row adds to the one above)	75 mM	150 mM
ESMC embedding, nine layers combined	18.0 %	20.4 %

Transmembrane helices resist salt. Of the 100 sparse-autoencoder features most associated with the salt response, 94 describe membrane helices (“transmembrane helix exit signature”, “multipass membrane helices”; 13 % of random features; $p = 6\text{e-}58$), all on the side of proteins whose HSPB1 binding survives or grows with salt. Proteins with a transmembrane helix gain 0.46 log2 relative to others; long disordered regions go the other way. This is what salt would do to the two kinds of interaction: binding through hydrophobic helices strengthens, binding through charged disordered regions weakens.

114 proteins leave the HSPB1 pull-down entirely at 150 mM (5 enter). A model trained without them or their relatives predicts them to fall (AUC 0.66 against proteins of matched abundance).

4. Heat

Raising the temperature of the lysate from 35 to 43 °C changes co-chaperone binding about 2.2-fold for a typical protein, and overall the pull-downs contain about 1.9 times more protein (run medians +0.9 log2), which with equal inputs means more total binding at 43 °C. The normalised responses below describe how each protein’s share of the binding changes.

Heat response	43 vs 35 °C	37 vs 35 °C
Amino-acid composition	8.9 %	3.5 %
ESMC, one layer (final)	25.7 %	11.4 %
ESMC, nine layers combined	27.4 %	12.7 %

Folded cytoplasmic proteins are recruited; disordered and membrane proteins are lost. The features that rise are those of folded enzyme cores (catalase, isocitrate and aldehyde dehydrogenases, enolase, TIM barrels, Rossmann folds; structural-motif features over-represented, odds ratio 5.5). Those that fall describe long disordered and low-complexity regions and transmembrane segments (membrane features over-represented, odds ratio 11.6).

A protein’s starting level matters. Proteins plentiful in the 35 °C pull-down tend to lose share at 43 °C, and scarce ones to gain (correlation -0.43; the starting level explains 16.5 % of the response). This is not shared measurement noise: taking the starting level and the response from different replicates gives the same correlation. It is regression to the mean in the biological sense: the 35 °C and 43 °C binding profiles agree only partly ($r = 0.6$), so proteins that dominate one tend to dominate the other less. In vitro there is also a direct reason: a fixed amount of chaperone means that new clients at 43 °C compete with the old. Sequence does not predict the response through the starting level: with the level removed, the embedding explains more of what is left (28.9 %).

4.1 Thermal stability acts through a window

If heat recruits proteins as they begin to unfold, the less stable a protein, the more it should gain. Melting temperatures for 9,792 human proteins come from the Meltome atlas (Jarzab et al. 2020; thermal proteome profiling, consensus over ten cell lines, reliability 0.96); 87 % of the heat-response proteins have one.

At 43 °C the relation is an inverted U. Proteins were sorted by melting temperature into ten equal groups:

Melting temperature (°C)	<46.7	-47.9	-48.9	-49.9	-51.0	-52.2	-53.6	-55.5	-58.4	>58.4
Change at 43 vs 35 °C	+0.11	+0.24	+0.41	+0.38	+0.37	+0.34	+0.27	+0.21	+0.10	+0.02
Change at 37 vs 35 °C	+0.07	+0.07	+0.08	+0.03	+0.03	+0.02	+0.01	-0.01	-0.03	-0.04

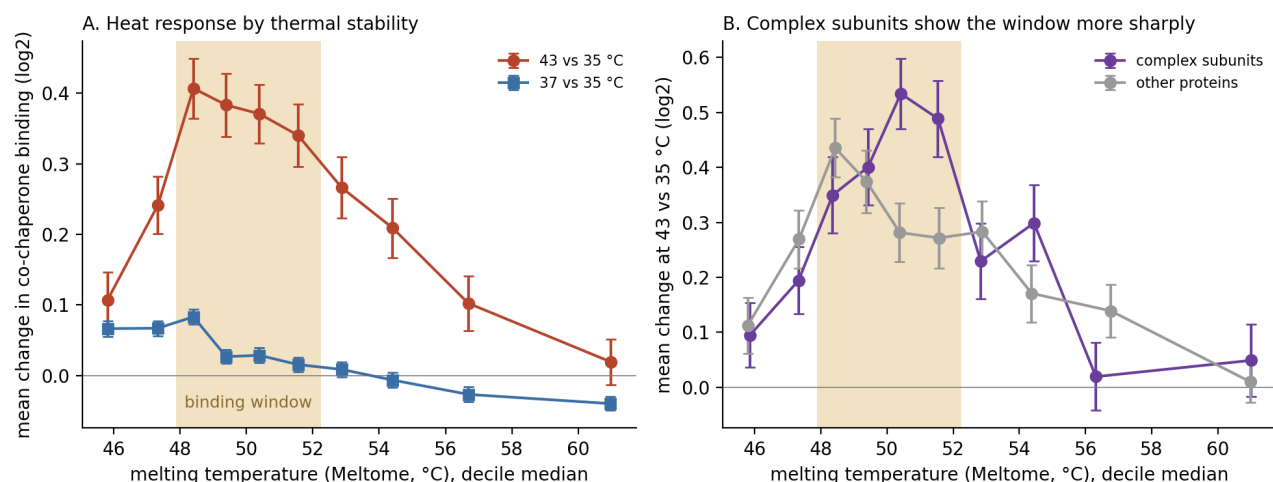


Figure 1: Mean change in each protein’s share of co-chaperone binding, by melting temperature (normalised; in absolute terms every group gains, see the table below). A: at 43 °C the gain rises with decreasing stability down to a melting temperature of about 48 °C, then falls for the least stable proteins, which responded earlier (shaded: the binding window); at 37 °C it rises steadily as stability decreases. B: at 43 °C, complex subunits (purple) show the window more sharply than other proteins (grey). Error bars, one standard error.

Reading the 43 °C curve from right to left: the most stable proteins (melting above 58 °C) are hardly affected by 43 °C and gain nothing (+0.02). As stability decreases, the gain grows, and it peaks for proteins that melt at 48-52 °C (+0.41), 5-8 °C above the incubation temperature. For the least stable proteins, melting below 47 °C, the gain falls again (+0.11). So the relation rises and then falls, and a straight-line correlation, which averages the two halves, is close to zero ($\rho = -0.04$). The curvature is statistically clear (quadratic term -0.073 [-0.095, -0.050], peak at 51 °C), appears with melting temperatures from HEK293T, K562 or Jurkat alone, and survives adjustment for protein class and starting level. At 37 °C, where nothing is near its melting point, the relation is the simple one: the least stable proteins gain most (slope -0.038 [-0.046, -0.028]).

Each protein’s binding switches on over a temperature range set by its stability. Two facts explain the inverted U. First, it describes each protein’s *share* of the binding: total binding nearly doubles at 43 °C, and in absolute terms every stability group gains (by +0.9 to +1.3 log2; table below), the least stable included. Second, splitting the 35 to 43 °C change into its two steps shows that proteins respond at different temperatures according to their stability:

Melting temperature (°C)	<46.7	-47.9	-48.9	-49.9	-51.0	-52.2	-53.6	-55.5	-58.4	>58.4
Step 35 to 37 °C	+0.07	+0.07	+0.08	+0.03	+0.03	+0.02	+0.01	-0.01	-0.02	-0.04
Step 37 to 43 °C	+0.04	+0.18	+0.32	+0.36	+0.34	+0.33	+0.26	+0.22	+0.13	+0.06
35 to 43 °C, absolute	+0.99	+1.12	+1.28	+1.26	+1.24	+1.22	+1.14	+1.09	+0.98	+0.89

Mean change, log2, for the 7,115 proteins detected at all three temperatures. “Absolute” adds back the overall rise removed by normalisation.

The least stable proteins gain most between 35 and 37 °C and then hardly change between 37 and 43 °C (+0.04, against +0.34 for proteins melting at 48-52 °C; $p = 2e-15$). Proteins of intermediate stability barely move up to 37 °C and respond strongly between 37 and 43 °C. The most stable have hardly begun by 43 °C. Each protein’s binding thus rises over a temperature range that tracks its stability, and comparing 43 with 35 °C catches each protein at a different point of its own transition: the least stable have already finished, the intermediate ones are mid-transition, and the most stable have not started. The least stable proteins did not start out more bound (their level in the 35 °C pull-down matches the others’), and western blots show the soluble proteins do not change across temperature, so the downturn reflects when binding happens, not loss of protein from solution. Soluble aggregates that do not pellet would still appear soluble on a blot, so that alternative is not entirely excluded.

But stability explains little of the heat response. Melting temperature, with its curvature, explains 1.2 % of the 43 °C response and adds 0.8 points to the sequence model; the sequence model’s predictions do not correlate with melting temperature at all ($r = 0.01$). The protein classes above (folded cytoplasmic versus disordered or membrane) act independently of stability. Proteins that the Meltome atlas could not melt, mostly disordered

Membership accounts for under 1 % of the response, like melting temperature: both point to real mechanisms, neither is the main driver.

4.3 Proteins that appear at 43 °C

131 proteins are detected in both co-chaperone pull-downs only at 43 °C, 16 times more than chance would give. They are the extreme of the same response: a model trained without them or their relatives predicts them to rise (AUC 0.70 against proteins of matched intensity; 0.83 for the strictest definition), they are cytoplasmic and rarely membrane proteins, and their melting temperature is ordinary (median 51 °C). They are listed as candidate heat-recruited clients in `reports/additional/heat43_appearing_proteins.tsv`.

5. Staurosporine

10 μ M staurosporine changes co-chaperone binding only about 1.15-fold, identically for DNAJA1 and DNAJB11. Sequence explains 4-7 % of it, under a tenth of the ceiling, and almost all of what it explains is about one class of proteins.

Staurosporine removes protein kinases. Kinases are 3.6 % of the proteins but carry 44-63 % of the variance of the response; other ATP-binding proteins do not move, and the heat response shows no kinase signal.

	Kinases	Other ATP-binding	All others	Kinase shift	p
35 °C	-0.15	+0.01	+0.00	-0.73 sd	2e-7
37 °C	-0.19	-0.01	+0.02	-1.05 sd	2e-20
43 °C	-0.20	-0.01	+0.00	-1.12 sd	5e-13

Depletion follows drug binding. Staurosporine dissociation constants for 374 kinases (ChEMBL; three kinome-wide studies agreeing at Spearman 0.93-0.98) show that tighter binders lose more, within kinase groups as well as between them (partial correlation +0.24 [+0.13, +0.34] at 43 °C). Leaving aside the AGC group (below), the mean change at 37 °C falls off steadily with affinity:

Dissociation constant	< 1 nM	1-10 nM	10-100 nM	0.1-1 μ M	1-10 μ M	> 10 μ M (not bound)
Mean change	-0.82	-0.24	-0.29	-0.22	-0.13	-0.04

Since staurosporine acts in the lysate, there is no cellular turnover: these are changes in binding (or in solubility). The reading that fits is that the drug, sitting in the ATP site, stabilises the kinase fold, and a stabilised kinase is no longer held by the co-chaperone.

Not through HSP90. Kinome-wide HSP90 client data (Taipale et al. 2012; 211 of these kinases) show that the kinases that lose most are *not* HSP90 clients: non-clients fall most (-0.29 to -0.37) and strong clients least (-0.03 to -0.12), and stronger HSP90 binding predicts a smaller loss (rho +0.24, p = 6e-4 at 43 °C). In the lysate, strong HSP90 clients may remain held by the endogenous HSP90-CDC37 machinery rather than by the added co-chaperone, and so have less to lose.

The AGC exception. The PKC isoforms, PKN and PKG bind staurosporine as tightly as any kinase (dissociation constants 0.2-50 nM) yet *gain* binding (PKG1 +4.4, PKCtheta +2.2, PKN2 +1.5). The sparse-autoencoder features that separate them mark AGC kinases with lipid- or cyclic-nucleotide-sensing regulatory domains (PKC, PKN, PKD, PKG) on the gaining side, and, on the losing side, CAMK2, DAPK and the STE kinases, through features labelled as disordered acidic tails, basic regulatory linkers and coiled-coil segments. These features separate rising from falling kinases well within known subfamilies (38 % of the variance at 43 °C, against 20 % for the embedding), but not for a kinase group never seen in training: direction is a subfamily property.

6. How the two co-chaperones differ

Which of the two co-chaperones a protein prefers is the most reproducible (96 %) and the most predictable response here: 49 % from sequence, a third from composition alone. Because both baits met the same lysate, the preference is intrinsic binding specificity.

Prefers DNAJB11		Prefers DNAJA1	
Signal peptide	+0.35 (p 6e-88)	SR splicing factors	-0.62
Transmembrane	+0.32 (p 1e-157)	Ribosomal proteins	-0.45
Glycoprotein	+0.31	Ribonucleoproteins	-0.42 (p 3e-58)
Disulfide bond	+0.29	KRAB zinc fingers	-0.46
Lysosome, Golgi, ER	+0.17 to +0.21	Nucleus	-0.15 (p 1e-158)

Mean preference, log2; > 0 toward DNAJB11.

DNAJB11 binds secretory and membrane proteins, the clients it meets in the ER, even when offered the whole proteome. DNAJA1 binds nuclear RNA-processing proteins; its sparse-autoencoder features are all charge and disorder features (“charged disordered low-complexity tracts”, “KR-rich NLS-like motifs”, “RS/SR phospho-regulated regions”). Family averages account for 60 % of what the model explains, and within families prediction still correlates at 0.56.

Different preferences, the same responses. Two different quantities are involved, and they behave differently.

- *Preference* is how much of a protein one co-chaperone holds relative to the other. The two pull-downs do not contain separate sets of proteins: 8,265 proteins are measured in both at 35 °C, and they differ in proportion, typically by 1.3- to 1.5-fold. The preferences above are tendencies of this kind, not exclusive client lists.
- *Response* is how a protein’s binding changes with a condition, measured separately in each pull-down. Across proteins, the heat response measured with DNAJA1 and the heat response measured with DNAJB11 correlate at 0.93; for staurosporine the two agree completely once measurement noise is accounted for (0.83-1.0). A protein that gains binding with heat gains it with both co-chaperones, and one that loses, loses with both. The response is therefore a property of the protein, of how it behaves when the lysate is heated or treated, rather than of the co-chaperone that captures it.

The two co-chaperones thus differ in *which proteins they favour*, but heat and staurosporine change those proteins in the same way for both. The one systematic difference is that heat weakens the preferences. The spread of preferences across proteins falls from 0.54 log2 at 35 °C to 0.36 at 43 °C, and each protein keeps about 40 % of its 35 °C preference on the log scale: a protein 1.5-fold more abundant in the DNAJB11 pull-down at 35 °C is about 1.2-fold more abundant at 43 °C. Nothing reproducible remains beyond this uniform shrinkage. The two pull-downs converge as the temperature rises, as expected if the clients that heat recruits are bound similarly by both co-chaperones and dilute their original preferences.

7. Across experiments

The responses are largely independent. The salt experiment predicts at most 7 % of any GA_33 response and vice versa, and staurosporine and the co-chaperone preference share nothing with salt. The one cross-experiment link, salt against heat ($r = -0.24$), disappears (-0.01) once each protein’s starting level in the two pull-downs is accounted for: proteins relatively enriched in one pull-down lose share in it under that experiment’s perturbation. There is no single protein property behind all the responses; each has its own determinants.

8. What would test these readings next

- **One input lysate.** Since every reaction started from the same lysate, measuring it once separates a protein’s amount in the lysate from how strongly it binds, and turns the starting-level effect into a binding

measure.

- **Binding at more temperatures.** Pull-downs at, say, 39 and 41 °C would trace each protein's transition directly, and test whether its midpoint follows its stability.
- **Melting temperatures measured in this lysate.** The Meltome values come from other cells and a three-minute heat pulse; lysate-specific values would sharpen the window.

Notes on methods

- Folds group proteins at 30 % sequence identity (MMseqs2). Out-of-family intervals resample families.
- The salt workbook had been damaged by a find-and-replace (missing values had become zeros, and eight accessions were mangled); targets were rebuilt from the replicates and checked against the file's own statistics.
- Weighting proteins by their measurement noise was tested and does not help: noise is too small a part of the variance for the salt and heat responses.
- Detailed analyses are in **reports/additional/additional_text_1** to **_7**: thermal stability (1), kinases and staurosporine (2), cross-experiment structure (3), normalisation (4), proteins that appear or vanish (5), combining layers (6), complexes (7).